

Promethium

Pm (Pm I). Strongest classified transitions below 2 μm (NIST): 23 levels, 16 lines.

Pm + ionisation limit 45020.80 cm⁻¹ = 5.58187 eV (222.120 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || e r || J' \rangle |$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

